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JEE MAIN · QUESTION PAPER

JEE Main 2022 (Online) 25th June Evening Shift

JEE Main Question Paper with Answer Key

25 Jun 2022	90	18	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	22 questions
Mathematics	23 questions
Physics	27 questions
Out of syllabus	18 questions
Answer key	at the end

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Chemistry 22 questions

Q1. MCQ (Single Correct Answer)

The minimum energy that must be possessed by photons in order to produce the photoelectric effect with platinum metal is :

[Given : The threshold frequency of platinum is $1.3 \times 10^{15}\ s^{-1}$ and h = 6.6×10^{-34} J s.]

(A) 3.21×10^{-14} J (B) 6.24×10^{-16} J (C) 8.58×10^{-19} J (D) 9.76×10^{-20} J

Open solution

Q2. MCQ (Single Correct Answer)

At 25°C and 1 atm pressure, the enthalpy of combustion of benzene (l) and acetylene (g) are $-3268\ kJ\ mol^{-1}$ and $-1300\ kJ\ mol^{-1}$, respectively. The change in enthalpy for the reaction 3 $C_2H_2(g) \rightarrow C_6H_6$ (l), is :

(A) $+324\ kJ\ mol^{-1}$ (B) $+632\ kJ\ mol^{-1}$ (C) $-632\ kJ\ mol^{-1}$ (D) $-732\ kJ\ mol^{-1}$

Open solution

Q3. MCQ (Single Correct Answer)

Solute A associates in water. When 0.7 g of solute A is dissolved in 42.0 g of water, it depresses the freezing point by 0.2°C. The percentage association of solute A in water, is :

[Given : Molar mass of A = 93 $g\ mol^{-1}$. Molal depression constant of water is 1.86 K kg mol^{-1}.]

(A) 50% (B) 60% (C) 70% (D) 80%

Open solution

Q4. MCQ (Single Correct Answer)

The K_{sp} for bismuth sulphide (Bi_2S_3) is 1.08×10^{-73}. The solubility of Bi_2S_3 in mol L^{-1} at 298 K is :

(A) 1.0×10^{-15} (B) 2.7×10^{-12} (C) 3.2×10^{-10} (D) 4.2×10^{-8}

Open solution

Q5. MCQ (Single Correct Answer)

The correct order of electron gain enthalpies of Cl, F, Te and Po is

(A) F < Cl < Te < Po (B) Po < Te < F < Cl (C) Te < Po < Cl < F (D) Cl < F < Te < Po

Open solution

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Q6. MCQ (Single Correct Answer)

Open solution

Given below are two statements one is labelled as Assertion A and the other is labelled as Reason R :

Assertion A : The amphoteric nature of water is explained by using Lewis acid/base concept.

Reason R : Water acts as an acid with NH_3 and as a base with H_2S.

In the light of the above statements choose the correct answer from the options given below :

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Q7. MCQ (Single Correct Answer)

Open solution

The correct order of reduction potentials of the following pairs is

A. Cl_2/Cl^-

B. I_2/I^-

C. Ag^+/Ag

D. Na^+/Na

E. Li^+/Li

Choose the correct answer from the options given below.

(A) A > C > B > D > E

(B) A > B > C > D > E

(C) A > C > B > E > D

(D) A > B > C > E > D

Q8. MCQ (Single Correct Answer)

Open solution

The metal ion (in gaseous state) with lowest spin-only magnetic moment value is :

(A) V^{2+} **(B)** Ni^{2+} **(C)** Cr^{2+} **(D)** Fe^{2+}

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Q9. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : A mixture contains benzoic acid and napthalene. The pure benzoic acid can be separated out by the use of benzene.

Reason R : Benzoic acid is soluble in hot water.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Q10. MCQ (Single Correct Answer)

Open solution

During halogen test, sodium fusion extract is boiled with concentrated HNO_3 to

(A) remove unreacted sodium

(B) decompose cyanide or sulphide of sodium

(C) extract halogen from organic compound

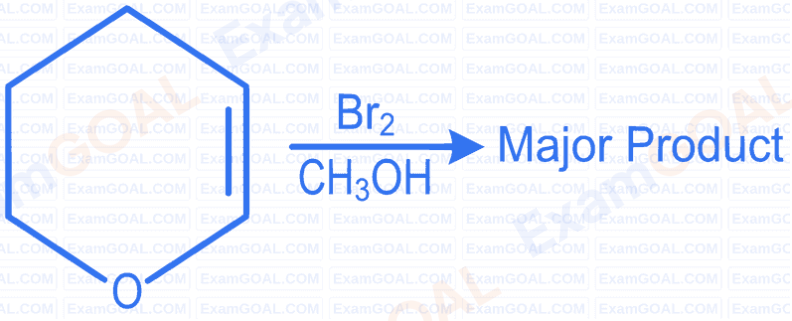
(D) maintain the pH of extract.

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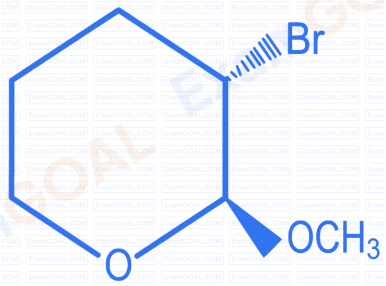
Q11. MCQ (Single Correct Answer)

Open solution

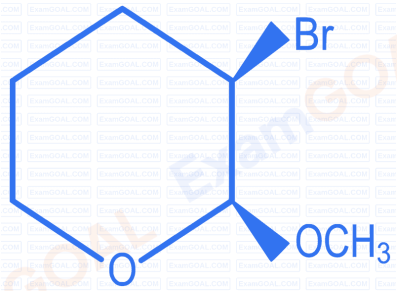
Amongst the following, the major product of the given chemical reaction is



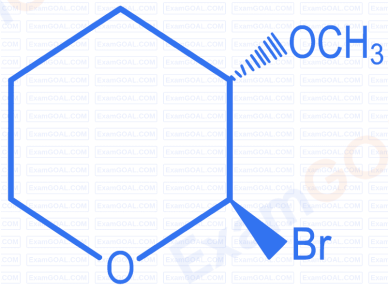
(A)



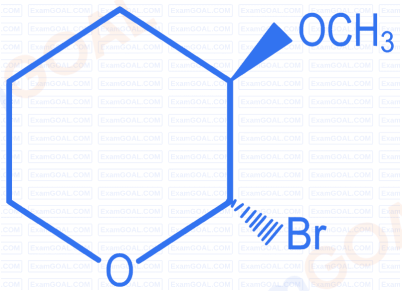
(B)



(C)



(D)



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Q12. MCQ (Single Correct Answer)

Open solution

In the given reaction



'A' can be

(A) benzyl bromide

(B) bromobenzene

(C) cyclohexyl bromide

(D) methyl bromide

Q13. MCQ (Single Correct Answer)

Open solution

Which of the following conditions or reaction sequence will NOT give acetophenone as the major product?

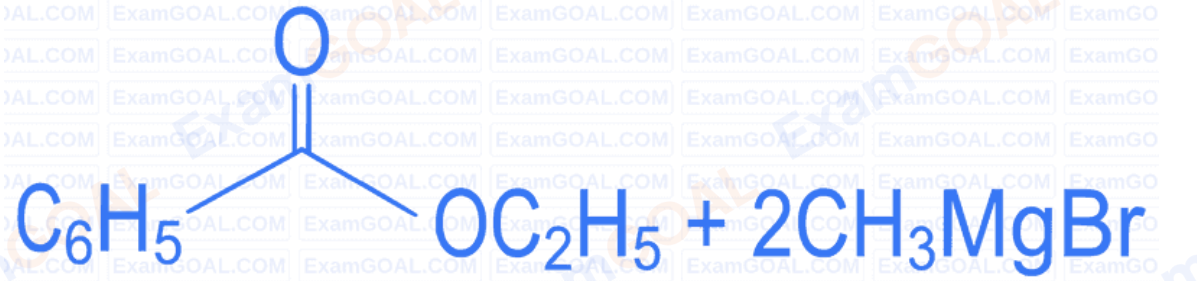
(A)



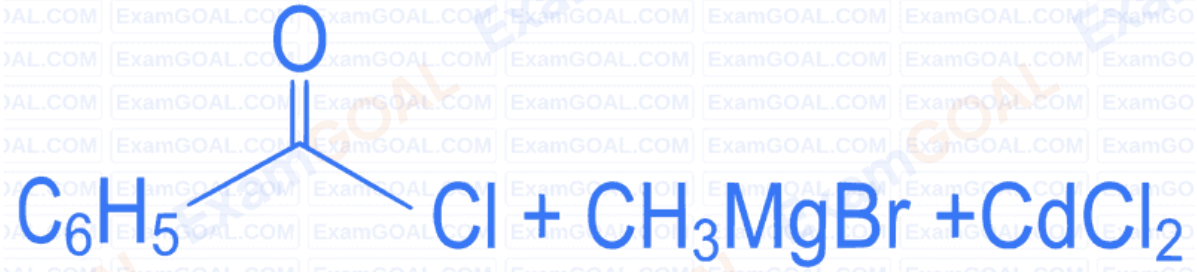
(B)



(C)



(D)



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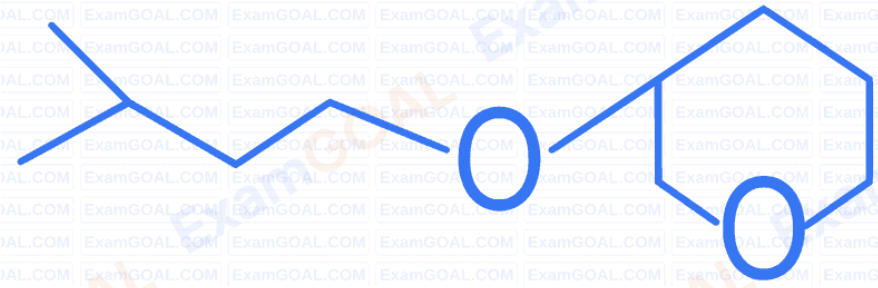
Q14. MCQ (Single Correct Answer)

Open solution

The major product formed in the following reaction, is



(A)



(B)



(C)



(D)



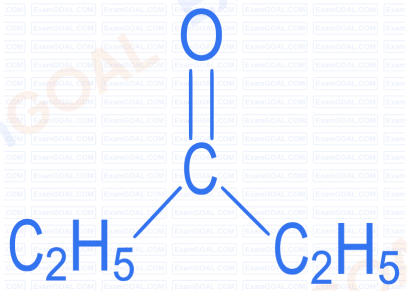
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Q15. MCQ (Single Correct Answer)

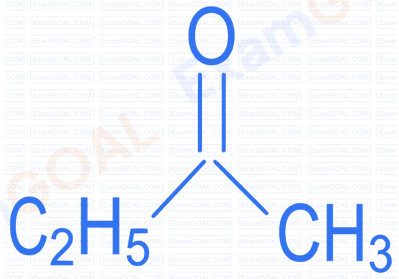
Which of the following ketone will NOT give enamine on treatment with secondary amines? [where t-Bu is $-C(CH_3)_3$]

Open solution

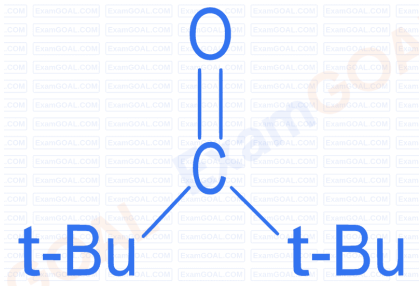
(A)



(B)



(C)



(D)



Q16. Numerical

A protein 'A' contains 0.30% of glycine (molecular weight 75). The minimum molar mass of the protein 'A' is _____ $\times 10^3$ g mol^{-1} [nearest integer]

Open solution

Q17. Numerical

Amongst BeF_2, BF_3, H_2O, NH_3, CCl_4 and HCl, the number of molecules with non-zero net dipole moment is _____.

Open solution

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Q18. Numerical

At 345 K, the half life for the decomposition of a sample of a gaseous compound initially at 55.5 kPa was 340 s. When the pressure was 27.8 kPa, the half life was found to be 170 s. The order of the reaction is _____. [integer answer]

Open solution

Q19. Numerical

A solution of $Fe_2(SO_4)_3$ is electrolyzed for 'x' min with a current of 1.5 A to deposit 0.3482 g of Fe. The value of x is _____. [nearest integer]

Given : 1 F = 96500 C mol^{-1}

Atomic mass of Fe = 56 g mol^{-1}

Open solution

Q20. Numerical

Amongst $FeCl_3.3H_2O$, $K_3[Fe(CN)_6]$ and $[Co(NH_3)_6]Cl_3$, the spin-only magnetic moment value of the inner-orbital complex that absorbs light at shortest wavelength is _____ B.M. [nearest integer]

Open solution

Q21. Numerical

How many of the given compounds will give a positive Biuret test _____ ?

Glycine, Glycylalanine, Tripeptide, Biuret

Open solution

Q22. Numerical

The neutralization occurs when 10 mL of 0.1M acid 'A' is allowed to react with 30 mL of 0.05 M base $M(OH)_2$. The basicity of the acid 'A' is _____.

[M is a metal]

Open solution

Mathematics 23 questions

Q23. MCQ (Single Correct Answer)

Let $A = \{x \in R : |x + 1| < 2\}$ and $B = \{x \in R : |x - 1| \geq 2\}$. Then which one of the following statements is NOT true?

(A) $A - B = (-1, 1)$

(B) $B - A = R - (-3, 1)$

(C) $A \cap B = (-3, -1]$

(D) $A \cup B = R - [1, 3)$

Open solution

Q24. MCQ (Single Correct Answer)

Let a, b $\in$ R be such that the equation $ax^2 - 2bx + 15 = 0$ has a repeated root α. If α and β are the roots of the equation $x^2 - 2bx + 21 = 0$, then $\alpha^2 + \beta^2$ is equal to :

(A) 37 (B) 58 (C) 68 (D) 92

Open solution

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Q25. MCQ (Single Correct Answer)

Let z_1 and z_2 be two complex numbers such that $\overline{z}_1 = i\overline{z}_2$ and $\arg\left(\frac{z_1}{\overline{z}_2}\right) = \pi$. Then :

(A) $\arg z_2 = \frac{\pi}{4}$ (B) $\arg z_2 = -\frac{3\pi}{4}$

(C) $\arg z_1 = \frac{\pi}{4}$ (D) $\arg z_1 = -\frac{3\pi}{4}$

Open solution

Q26. MCQ (Single Correct Answer)

The system of equations

$-kx + 3y - 14z = 25$

$-15x + 4y - kz = 3$

$-4x + y + 3z = 4$

is consistent for all k in the set

(A) R (B) $R - \{-11, 13\}$ (C) $R - \{13\}$ (D) $R - \{-11, 11\}$

Open solution

Q27. MCQ (Single Correct Answer)

$\lim\limits_{x \to \frac{\pi}{2}} \left(\tan^2 x \left((2\sin^2 x + 3\sin x + 4)^{\frac{1}{2}} - (\sin^2 x + 6\sin x + 2)^{\frac{1}{2}}\right)\right)$ is equal to

(A) $\frac{1}{12}$ (B) $-\frac{1}{18}$ (C) $-\frac{1}{12}$ (D) $\frac{1}{6}$

Open solution

Q28. MCQ (Single Correct Answer)

The area of the region enclosed between the parabolas $y^2 = 2x - 1$ and $y^2 = 4x - 3$ is

(A) $\frac{1}{3}$ (B) $\frac{1}{6}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$

Open solution

Q29. MCQ (Single Correct Answer)

The coefficient of x^{101} in the expression $(5+x)^{500} + x(5+x)^{499} + x^2(5+x)^{498} + \ldots\ldots + x^{500}$, x > 0, is

(A) ${}^{501}C_{101}\,(5)^{399}$ (B) ${}^{501}C_{101}\,(5)^{400}$ (C) ${}^{501}C_{100}\,(5)^{400}$ (D) ${}^{500}C_{101}\,(5)^{399}$

Open solution

Q30. MCQ (Single Correct Answer)

Water is being filled at the rate of 1 cm^3 / sec in a right circular conical vessel (vertex downwards) of height 35 cm and diameter 14 cm. When the height of the water level is 10 cm, the rate (in cm^2 / sec) at which the wet conical surface area of the vessel increases is

(A) 5 (B) $\frac{\sqrt{21}}{5}$ (C) $\frac{\sqrt{26}}{5}$ (D) $\frac{\sqrt{26}}{10}$

Open solution

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Q31. MCQ (Single Correct Answer)

If $b_n = \int_0^{\frac{\pi}{2}} \frac{\cos^2 nx}{\sin x} dx$, $n \in N$, then

(A) $b_3 - b_2$, $b_4 - b_3$, $b_5 - b_4$ are in A.P. with common difference -2

(B) $\frac{1}{b_3 - b_2}$, $\frac{1}{b_4 - b_3}$, $\frac{1}{b_5 - b_4}$ are in an A.P. with common difference 2

(C) $b_3 - b_2$, $b_4 - b_3$, $b_5 - b_4$ are in a G.P.

(D) $\frac{1}{b_3 - b_2}$, $\frac{1}{b_4 - b_3}$, $\frac{1}{b_5 - b_4}$ are in an A.P. with common difference -2

Open solution

Q32. MCQ (Single Correct Answer)

If $y = y(x)$ is the solution of the differential equation

$2x^2 \frac{dy}{dx} - 2xy + 3y^2 = 0$ such that $y(e) = \frac{e}{3}$, then y(1) is equal to :

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) 3

Open solution

Q33. MCQ (Single Correct Answer)

The value of 2sin (12°) − sin (72°) is :

(A) $\frac{\sqrt{5}(1-\sqrt{3})}{4}$ (B) $\frac{1-\sqrt{5}}{8}$

(C) $\frac{\sqrt{3}(1-\sqrt{5})}{2}$ (D) $\frac{\sqrt{3}(1-\sqrt{5})}{4}$

Open solution

Q34. MCQ (Single Correct Answer)

A biased die is marked with numbers 2, 4, 8, 16, 32, 32 on its faces and the probability of getting a face with mark n is $\frac{1}{n}$. If the die is thrown thrice, then the probability, that the sum of the numbers obtained is 48, is :

(A) $\frac{7}{2^{11}}$ (B) $\frac{7}{2^{12}}$ (C) $\frac{3}{2^{10}}$ (D) $\frac{13}{2^{12}}$

Open solution

Q35. MCQ (Single Correct Answer)

The value of $\tan^{-1}\left(\frac{\cos\left(\frac{15\pi}{4}\right)-1}{\sin\left(\frac{\pi}{4}\right)}\right)$ is equal to :

(A) $-\frac{\pi}{4}$ (B) $-\frac{\pi}{8}$ (C) $-\frac{5\pi}{12}$ (D) $-\frac{4\pi}{9}$

Open solution

Q36. MCQ (Single Correct Answer)

The line y = x + 1 meets the ellipse $\frac{x^2}{4} + \frac{y^2}{2} = 1$ at two points P and Q. If r is the radius of the circle with PQ as diameter then $(3r)^2$ is equal to :

(A) 20 (B) 12 (C) 11 (D) 8

Open solution

Q37. Numerical

Let $A = \begin{pmatrix} 2 & -2 \\ 1 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} -1 & 2 \\ -1 & 2 \end{pmatrix}$. Then the number of elements in the set {(n, m) : n, m ∈ {1, 2, ..., 10} and $nA^n + mB^m = I$} is _____.

Open solution

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Q38. Numerical

Let $f(x) = [2x^2 + 1]$ and $g(x) = \begin{cases} 2x - 3, & x < 0 \\ 2x + 3, & x \geq 0 \end{cases}$, where [t] is the greatest integer $\leq$ t. Then, in the open interval $(-1, 1)$, the number of points where fog is discontinuous is equal to _____.

Open solution

Q39. Numerical

The value of b > 3 for which $12\int_3^b \frac{1}{(x^2-1)(x^2-4)}dx = \log_e\left(\frac{49}{40}\right)$, is equal to _____.

Open solution

Q40. Numerical

If the sum of the co-efficient of all the positive even powers of x in the binomial expansion of $\left(2x^3 + \frac{3}{x}\right)^{10}$ is $5^{10} - \beta \cdot 3^9$, then β is equal to _____.

Open solution

Q41. Numerical

If the mean deviation about the mean of the numbers 1, 2, 3, ..., n, where n is odd, is $\frac{5(n+1)}{n}$, then n is equal to _____.

Open solution

Q42. Numerical

Let $\vec{b} = \hat{i} + \hat{j} + \lambda\hat{k}$, $\lambda \in$ R. If $\vec{a}$ is a vector such that $\vec{a} \times \vec{b} = 13\hat{i} - \hat{j} - 4\hat{k}$ and $\vec{a} \cdot \vec{b} + 21 = 0$, then $\left(\vec{b} - \vec{a}\right) \cdot \left(\hat{k} - \hat{j}\right) + \left(\vec{b} + \vec{a}\right) \cdot \left(\hat{i} - \hat{k}\right)$ is equal to _____.

Open solution

Q43. Numerical

The total number of three-digit numbers, with one digit repeated exactly two times, is _____.

Open solution

Q44. Numerical

Let $f(x) = |(x-1)(x^2 - 2x - 3)| + x - 3$, $x \in R$. If m and M are respectively the number of points of local minimum and local maximum of f in the interval (0, 4), then m + M is equal to _____.

Open solution

Q45. Numerical

Let l_1 be the line in xy-plane with x and y intercepts $\frac{1}{8}$ and $\frac{1}{4\sqrt{2}}$ respectively, and l_2 be the line in zx-plane with x and z intercepts $-\frac{1}{8}$ and $-\frac{1}{6\sqrt{3}}$ respectively. If d is the shortest distance between the line l_1 and l_2, then d^{-2} is equal to _____.

Open solution

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Physics 27 questions

Q46. MCQ (Single Correct Answer)

Given below are two statements. One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Two identical balls A and B thrown with same velocity 'u' at two different angles with horizontal attained the same range R. If A and B reached the maximum height h_1 and h_2 respectively, then $R = 4\sqrt{h_1 h_2}$

Reason R : Product of said heights.

$h_1 h_2 = \left(\frac{u^2 \sin^2\theta}{2g}\right) \cdot \left(\frac{u^2 \cos^2\theta}{2g}\right)$

Choose the correct answer :

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Open solution

Q47. MCQ (Single Correct Answer)

Two buses P and Q start from a point at the same time and move in a straight line and their positions are represented by $X_P(t) = \alpha t + \beta t^2$ and $X_Q(t) = ft - t^2$. At what time, both the buses have same velocity?

(A) $\frac{\alpha - f}{1+\beta}$ (B) $\frac{\alpha + f}{2(\beta - 1)}$

(C) $\frac{\alpha + f}{2(1+\beta)}$ (D) $\frac{f - \alpha}{2(1+\beta)}$

Open solution

Q48. MCQ (Single Correct Answer)

A disc with a flat small bottom beaker placed on it at a distance R from its center is revolving about an axis passing through the center and perpendicular to its plane with an angular velocity ω. The coefficient of static friction between the bottom of the beaker and the surface of the disc is μ. The beaker will revolve with the disc if :

(A) $R \leq \frac{\mu g}{2\omega^2}$ (B) $R \leq \frac{\mu g}{\omega^2}$

(C) $R \geq \frac{\mu g}{2\omega^2}$ (D) $R \geq \frac{\mu g}{\omega^2}$

Open solution

Q49. MCQ (Single Correct Answer)

A solid metallic cube having total surface area 24 m^2 is uniformly heated. If its temperature is increased by 10°C, calculate the increase in volume of the cube. (Given α = 5.0 $\times$ 10^{-4} $^\circ C^{-1}$).

(A) 2.4 $\times$ 10^6 cm^3 (B) 1.2 $\times$ 10^5 cm^3 (C) 6.0 $\times$ 10^4 cm^3 (D) 4.8 $\times$ 10^5 cm^3

Open solution

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Q50. MCQ (Single Correct Answer)

A copper block of mass 5.0 kg is heated to a temperature of 500°C and is placed on a large ice block. What is the maximum amount of ice that can melt? [Specific heat of copper : 0.39 J g^{-1} $^{\circ}C^{-1}$ and latent heat of fusion of water : 335 J g^{-1}]

Open solution

(A) 1.5 kg (B) 5.8 kg (C) 2.9 kg (D) 3.8 kg

Q51. MCQ (Single Correct Answer)

The ratio of specific heats $\left(\frac{C_P}{C_V}\right)$ in terms of degree of freedom (f) is given by :

Open solution

(A) $\left(1+\frac{f}{3}\right)$ (B) $\left(1+\frac{2}{f}\right)$ (C) $\left(1+\frac{f}{2}\right)$ (D) $\left(1+\frac{1}{f}\right)$

Q52. MCQ (Single Correct Answer)

For a particle in uniform circular motion, the acceleration $\vec{a}$ at any point P(R, θ) on the circular path of radius R is (when θ is measured from the positive x-axis and v is uniform speed) :

Open solution

(A) $-\frac{v^2}{R}\sin\theta\hat{i}+\frac{v^2}{R}\cos\theta\hat{j}$

(B) $-\frac{v^2}{R}\cos\theta\hat{i}+\frac{v^2}{R}\sin\theta\hat{j}$

(C) $-\frac{v^2}{R}\cos\theta\hat{i}-\frac{v^2}{R}\sin\theta\hat{j}$

(D) $-\frac{v^2}{R}\hat{i}+\frac{v^2}{R}\hat{j}$

Q53. MCQ (Single Correct Answer)

Two metallic plates form a parallel plate capacitor. The distance between the plates is 'd'. A metal sheet of thickness $\frac{d}{2}$ and of area equal to area of each plate is introduced between the plates. What will be the ratio of the new capacitance to the original capacitance of the capacitor?

Open solution

(A) 2 : 1 (B) 1 : 2 (C) 1 : 4 (D) 4 : 1

Q54. MCQ (Single Correct Answer)

Two cells of same emf but different internal resistances r_1 and r_2 are connected in series with a resistance R. The value of resistance R, for which the potential difference across second cell is zero, is :

Open solution

(A) $r_2 - r_1$ (B) $r_1 - r_2$ (C) r_1 (D) r_2

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Q55. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : Susceptibilities of paramagnetic and ferromagnetic substances increase with decrease in temperature.

Statement II : Diamagnetism is a result of orbital motions of electrons developing magnetic moments opposite to the applied magnetic field.

Choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is true but Statement II is false.

(D) Statement I is false but Statement II is true.

Q56. MCQ (Single Correct Answer)

Open solution

A long solenoid carrying a current produces a magnetic field B along its axis. If the current is doubled and the number of turns per cm is halved, the new value of magnetic field will be equal to

(A) B (B) 2B (C) 4B (D) $\frac{B}{2}$

Q57. MCQ (Single Correct Answer)

Open solution

A sinusoidal voltage V(t) = 210 sin 3000 t volt is applied to a series LCR circuit in which L = 10 mH, C = 25 μF and R = 100 Ω. The phase difference (Φ) between the applied voltage and resultant current will be :

(A) $\tan^{-1}(0.17)$ (B) $\tan^{-1}(9.46)$ (C) $\tan^{-1}(0.30)$ (D) $\tan^{-1}(13.33)$

Q58. MCQ (Single Correct Answer)

Open solution

The electromagnetic waves travel in a medium at a speed of 2.0×10^8 m/s. The relative permeability of the medium is 1.0. The relative permittivity of the medium will be :

(A) 2.25 (B) 4.25 (C) 6.25 (D) 8.25

Q59. MCQ (Single Correct Answer)

Open solution

The interference pattern is obtained with two coherent light sources of intensity ratio 4 : 1. And the ratio $\frac{I_{max}+I_{min}}{I_{max}-I_{min}}$ is $\frac{5}{x}$. Then, the value of x will be equal to :

(A) 3 (B) 4 (C) 2 (D) 1

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Q60. MCQ (Single Correct Answer)

A light whose electric field vectors are completely removed by using a good polaroid, allowed to incident on the surface of the prism at Brewster's angle. Choose the most suitable option for the phenomenon related to the prism.

Open solution

(A) Reflected and refracted rays will be perpendicular to each other.

(B) Wave will propagate along the surface of prism.

(C) No refraction, and there will be total reflection of light.

(D) No reflection, and there will be total transmission of light.

Q61. MCQ (Single Correct Answer)

A proton, a neutron, an electron and an α particle have same energy. If λ_p, λ_n, λ_e and λ_α are the de Broglie's wavelengths of proton, neutron, electron and α particle respectively, then choose the correct relation from the following :

Open solution

(A) $\lambda_p = \lambda_n > \lambda_e > \lambda_\alpha$

(B) $\lambda_\alpha < \lambda_n < \lambda_p < \lambda_e$

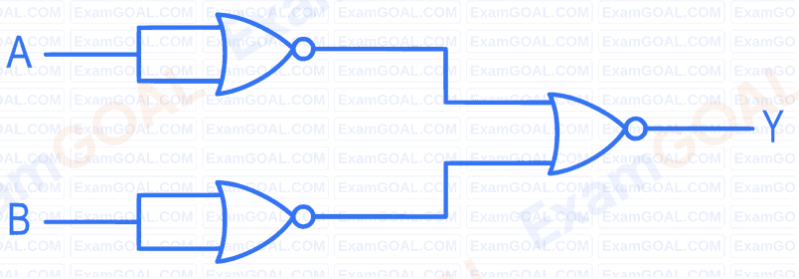
(C) $\lambda_e < \lambda_p = \lambda_n > \lambda_\alpha$

(D) $\lambda_e = \lambda_p = \lambda_n = \lambda_\alpha$

Q62. MCQ (Single Correct Answer)

Identify the logic operation performed by the given circuit:

Open solution



(A) AND gate

(B) OR gate

(C) NOR gate

(D) NAND gate

Q63. MCQ (Single Correct Answer)

If n represents the actual number of deflections in a converted galvanometer of resistance G and shunt resistance S. Then the total current I when its figure of merit is K will be:

Open solution

(A) $\frac{KS}{(S+G)}$

(B) $\frac{(G+S)}{nKS}$

(C) $\frac{nKS}{(G+S)}$

(D) $\frac{nK(G+S)}{S}$

Q64. Numerical

For $z = a^2x^3y^{\frac{1}{2}}$, where 'a' is a constant. If percentage error in measurement of 'x' and 'y' are 4% and 12% respectively, then the percentage error for 'z' will be _____%.

Open solution

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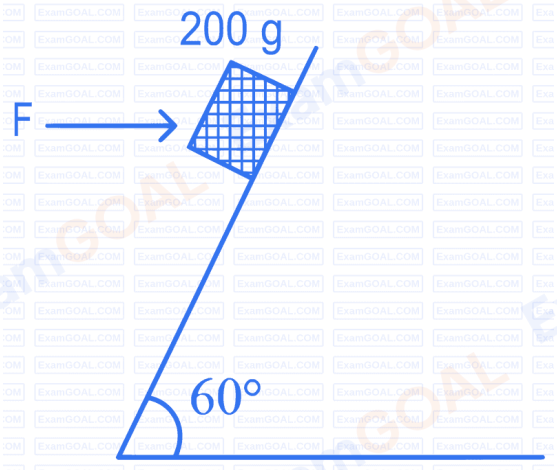
Q65. Numerical

A curve in a level road has a radius 75 m. The maximum speed of a car turning this curved road can be 30 m/s without skidding. If radius of curved road is changed to 48 m and the coefficient of friction between the tyres and the road remains same, then maximum allowed speed would be _____ m/s.

Open solution

Q66. Numerical

A block of mass 200 g is kept stationary on a smooth inclined plane by applying a minimum horizontal force F = $\sqrt{x}$N as shown in figure.



The value of x = _____.

Open solution

Q67. Numerical

Moment of Inertia (M.I.) of four bodies having same mass 'M' and radius '2R' are as follows:

I_1 = M.I. of solid sphere about its diameter

I_2 = M.I. of solid cylinder about its axis

I_3 = M.I. of solid circular disc about its diameter

I_4 = M.I. of thin circular ring about its diameter

If $2(I_2 + I_3) + I_4 = x \cdot I_1$, then the value of x will be _____.

Open solution

Q68. Numerical

Two satellites S_1 and S_2 are revolving in circular orbits around a planet with radius R_1 = 3200 km and R_2 = 800 km respectively. The ratio of speed of satellite S_1 to be speed of satellite S_2 in their respective orbits would be $\frac{1}{x}$ where x = _____.

Open solution

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Q69. Numerical

When a gas filled in a closed vessel is heated by raising the temperature by 1°C, its pressure increases by 0.4%. The initial temperature of the gas is _____ K.

Open solution

Q70. Numerical

27 identical drops are charged at 22V each. They combine to form a bigger drop. The potential of the bigger drop will be _____ V.

Open solution

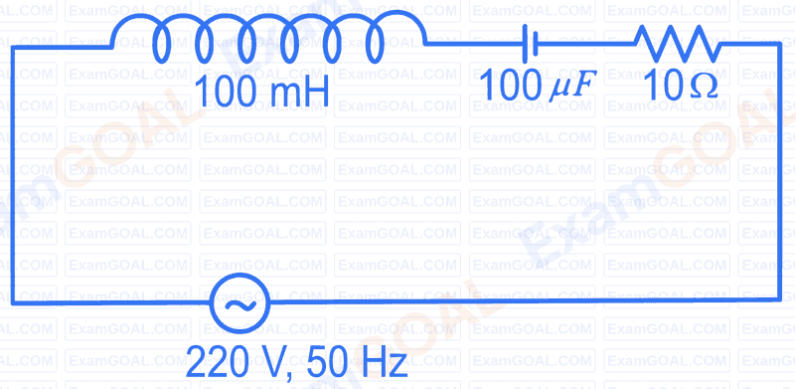
Q71. Numerical

The length of a given cylindrical wire is increased to double of its original length. The percentage increase in the resistance of the wire will be _____ %.

Open solution

Q72. Numerical

In a series LCR circuit, the inductance, capacitance and resistance are L = 100 mH, C = 100 μF and R = 10 Ω respectively. They are connected to an AC source of voltage 220 V and frequency of 50 Hz. The approximate value of current in the circuit will be _____ A.



Open solution

Out of syllabus 18 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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Q73. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List I with List II

	List - I		List -II
A.	Zymase	I.	Stomach
B.	Diastase	II.	Yeast
C.	Urease	III.	Malt
D.	Pepsin	IV.	Soyabean

Choose the correct answer from the options given below :

(A) A-II, B-III, C-I, D-IV

(B) A-II, B-III, C-IV, D-I

(C) A-III, B-II, C-IV, D-I

(D) A-III, B-II, C-I, D-IV

Q74. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements.

Statement I : During electrolytic refining, blister copper deposits precious metals.

Statement II : In the process of obtaining pure copper by electrolysis method, copper blister is used to make the anode.

In the light of the above statements, choose the correct from the options given below.

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is true but Statement II is false.

(D) Statement I is false but Statement II is true.

Q75. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The number of bridged oxygen atoms present in compound B formed from the following reactions is

$Pb(NO_3)_2 \xrightarrow{673\ K} A + PbO + O_2$

$A \xrightarrow{Dimerise} B$

(A) 0 **(B)** 1 **(C)** 2 **(D)** 3

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Q76. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Polluted water may have a value of BOD of the order of 17 ppm.

Reason R : BOD is a measure of oxygen required to oxidise both the bio-degradable and non-biodegradable organic material in water.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) Both A and R are correct and R is the correct explanation of A.

(B) Both A and R are correct but R is NOT the correct explanation of A.

(C) A is correct but R is not correct.

(D) A is not correct but R is correct.

Q77. MCQ (Single Correct Answer) · **Chemistry**

Open solution

An antiseptic dettol is a mixture of two compounds 'A' and 'B' where A has 6π electrons and B has 2π electrons. What is 'B'?

(A) Bithionol **(B)** Terpineol **(C)** Chloroxylenol **(D)** Chloramphenicol

Q78. Numerical · **Chemistry**

Open solution

A rigid nitrogen tank stored inside a laboratory has a pressure of 30 atm at 06:00 am when the temperature is 27°C. At 03:00 pm, when the temperature is 45°C, the pressure in the tank will be _____ atm. [nearest integer]

Q79. Numerical · **Chemistry**

Open solution

Consider the following reactions:

$PCl_3 + H_2O \rightarrow A + HCl$

$A + H_2O \rightarrow B + HCl$

The number of ionisable protons present in the product B is _____.

Q80. Numerical · **Chemistry**

Open solution

The Novolac polymer has mass of 963 g. The number of monomer units present in it are

Q81. MCQ (Single Correct Answer) · **Mathematics**

Open solution

The sum $1 + 2 \cdot 3 + 3 \cdot 3^2 + \ldots\ldots + 10 \cdot 3^9$ is equal to :

(A) $\frac{2 \cdot 3^{12}+10}{4}$ **(B)** $\frac{19 \cdot 3^{10}+1}{4}$ **(C)** $5 \cdot 3^{10} - 2$ **(D)** $\frac{9 \cdot 3^{10}+1}{2}$

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Q82. MCQ (Single Correct Answer) · **Mathematics**

Let p be the plane passing through the intersection of the planes $\vec{r}.\left(\hat{i}+3\hat{j}-\hat{k}\right)=5$ and $\vec{r}.\left(2\hat{i}-\hat{j}+\hat{k}\right)=3$, and the point (2, 1, −2). Let the position vectors of the points X and Y be $\hat{i}-2\hat{j}+4\hat{k}$ and $5\hat{i}-\hat{j}+2\hat{k}$ respectively. Then the points :

(A) X and X + Y are on the same side of P

(B) Y and Y − X are on the opposite sides of P

(C) X and Y are on the opposite sides of P

(D) X + Y and X − Y are on the same side of P

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

A circle touches both the y-axis and the line x + y = 0. Then the locus of its center is :

(A) $y=\sqrt{2}x$ (B) $x=\sqrt{2}y$ (C) $y^2-x^2=2xy$ (D) $x^2-y^2=2xy$

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

If the angle made by the tangent at the point (x_0, y_0) on the curve $x=12(t+\sin t\cos t)$, $y=12(1+\sin t)^2$, $0<t<\frac{\pi}{2}$, with the positive x-axis is $\frac{\pi}{3}$, then y_0 is equal to:

(A) $6\left(3+2\sqrt{2}\right)$ (B) $3\left(7+4\sqrt{3}\right)$ (C) 27 (D) 48

Open solution

Q85. MCQ (Single Correct Answer) · **Mathematics**

The negation of the Boolean expression $((\sim q)\wedge p)\Rightarrow((\sim p)\vee q)$ is logically equivalent to :

(A) $p\Rightarrow q$ (B) $q\Rightarrow p$ (C) $\sim(p\Rightarrow q)$ (D) $\sim(q\Rightarrow p)$

Open solution

Q86. MCQ (Single Correct Answer) · **Mathematics**

If the line $y=4+kx,\ k>0$, is the tangent to the parabola $y=x-x^2$ at the point P and V is the vertex of the parabola, then the slope of the line through P and V is :

(A) $\frac{3}{2}$ (B) $\frac{26}{9}$ (C) $\frac{5}{2}$ (D) $\frac{23}{6}$

Open solution

Q87. Numerical · **Mathematics**

Let the eccentricity of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ be $\frac{5}{4}$. If the equation of the normal at the point $\left(\frac{8}{\sqrt{5}},\frac{12}{5}\right)$ on the hyperbola is $8\sqrt{5}x+\beta y=\lambda$, then $\lambda-\beta$ is equal to _____.

Open solution

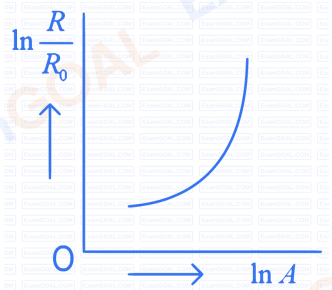
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Q88. MCQ (Single Correct Answer) · **Physics**

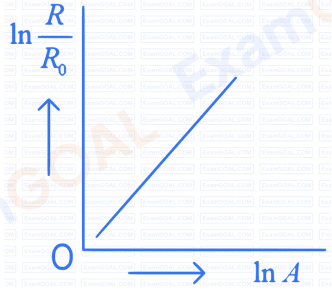
Which of the following figure represents the variation of $\ln\left(\frac{R}{R_0}\right)$ with $\ln A$ (if R = radius of a nucleus and A = its mass number)

Open solution

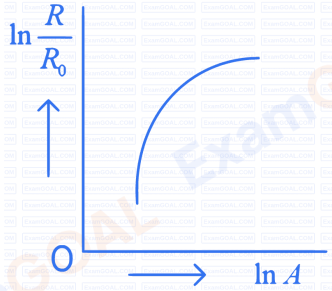
(A)



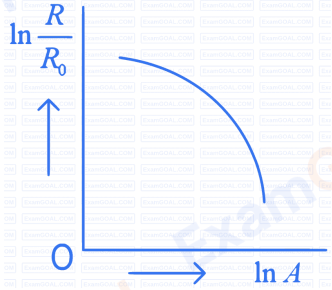
(B)



(C)



(D)



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Q89. MCQ (Single Correct Answer) · **Physics**

Open solution

Match List I with List II

	List - I		List -II
A.	Facsimile	I.	Static Document Image
B.	Guided media Channel	II.	Local Broadcast Radio
C.	Frequency Modulation	III.	Rectangular wave
D.	Digital Signal	IV.	Optical Fiber

Choose the correct answer from the following options:

(A) A-IV, B-III, C-II, D-I

(B) A-I, B-IV, C-II, D-III

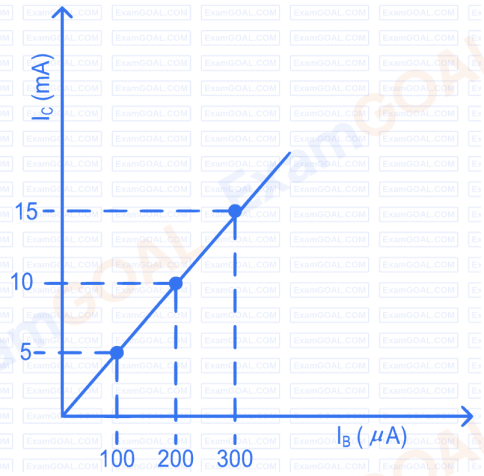
(C) A-IV, B-II, C-III, D-I

(D) A-I, B-II, C-III, D-IV

Q90. Numerical · **Physics**

Open solution

In an experiment of CE configuration of n-p-n transistor, the transfer characteristics are observed as given in figure.



If the input resistance is 200 Ω and output resistance is 60 Ω, the voltage gain in this experiment will be _____.

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Answer Key

Chemistry

1. **C**	2. **C**	3. **D**	4. **A**	5. **B**
6. **D**	7. **A**	8. **B**	9. **D**	10. **B**
11. **A**	12. **B**	13. **C**	14. **D**	15. **C**
16. **25**	17. **3**	18. **0**	19. **20**	20. **2**
21. **2**	22. **3**			

Mathematics

23. **B**	24. **B**	25. **C**	26. **D**	27. **A**
28. **A**	29. **A**	30. **C**	31. **D**	32. **B**
33. **D**	34. **D**	35. **B**	36. **A**	37. **1**
38. **62**	39. **6**	40. **83**	41. **21**	42. **14**
43. **243**	44. **3**	45. **51**		

Physics

46. **A**	47. **D**	48. **B**	49. **B**	50. **C**
51. **B**	52. **C**	53. **A**	54. **A**	55. **A**
56. **A**	57. **A**	58. **A**	59. **B**	60. **D**
61. **B**	62. **A**	63. **D**	64. **18**	65. **24**
66. **12**	67. **5**	68. **2**	69. **250**	70. **198**
71. **300**	72. **22**			

Out of syllabus

73. **B**	74. **A**	75. **A**	76. **C**	77. **B**
78. **32**	79. **2**	80. **9**	81. **B**	82. **C**
83. **D**	84. **C**	85. **C**	86. **C**	87. **85**
88. **B**	89. **B**	90. **15**		

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